BR2 Physical Breakout Map - Bank A

Schematic jack -> header pin -> Au/FPGA A-pin. This matches the 4 x 14 physical breakout (56 holes).

Editable PDF form

JACK	PIN	AU PIN	TYPE	USED	SIGNAL / PURPOSE	DIR	STATUS
J7	1	GND	GND				
J7	2	A3	IO N	☐			
J7	3	A9	IO N	☐			
J7	4	A15	IO N	☐			
J7	5	A21	IO N	☐			
J7	6	A27	IO N	☐			
J7	7	A33	IO N	☐			
J7	8	A39	IO N - GCLK	☐			
J7	9	A45	IO N - GCLK	☐			
J7	10	A51	IO N	☐			
J7	11	A57	IO N A	☐			
J7	12	A63	IO N A	☐			
J7	13	A69	IO N A	☐			
J7	14	A75	IO N A	☐			

JACK	PIN	AU PIN	TYPE	USED	SIGNAL / PURPOSE	DIR	STATUS
J8	1	GND	GND				
J8	2	A5	IO P	☐			
J8	3	A11	IO P	☐			
J8	4	A17	IO P	☐			
J8	5	A23	IO P	☐			
J8	6	A29	IO P	☐			
J8	7	A35	IO P	☐			
J8	8	A41	IO P - GCLK	☐			
J8	9	A47	IO P - GCLK	☐			
J8	10	A53	IO P	☐			
J8	11	A59	IO P A	☐			
J8	12	A65	IO P A	☐			
J8	13	A71	IO P A	☐			
J8	14	A77	IO P A	☐			

JACK	PIN	AU PIN	TYPE	USED	SIGNAL / PURPOSE	DIR	STATUS
J9	1	GND	GND				
J9	2	A4	IO N	☐			
J9	3	A10	IO N	☐			
J9	4	A16	IO N	☐			
J9	5	A22	IO N	☐			
J9	6	A28	IO N	☐			
J9	7	A34	IO N	☐			
J9	8	A40	IO N - LCLK	☐			
J9	9	A46	IO N - LCLK	☐			
J9	10	A52	IO N	☐			
J9	11	A58	IO N A	☐			
J9	12	A64	IO N A	☐			
J9	13	A70	IO N A	☐			
J9	14	A76	IO N A	☐			

JACK	PIN	AU PIN	TYPE	USED	SIGNAL / PURPOSE	DIR	STATUS
J10	1	GND	GND				
J10	2	A6	IO P	☐			
J10	3	A12	IO P	☐			
J10	4	A18	IO P	☐			
J10	5	A24	IO P	☐			
J10	6	A30	IO P	☐			
J10	7	A36	IO P	☐			
J10	8	A42	IO P - LCLK	☐			
J10	9	A48	IO P - LCLK	☐			
J10	10	A54	IO P	☐			
J10	11	A60	IO P A	☐			
J10	12	A66	IO P A	☐			
J10	13	A72	IO P A	☐			
J10	14	A78	IO P A	☐			

DIR: IN / OUT / BIDIR. STATUS: PLANNED / WIRED / PASS. 5 V logic into FPGA I/O requires level shifting.

BR2 Physical Breakout Map - Bank B

Schematic jack -> header pin -> Au/FPGA B-pin. This matches the 4 x 14 physical breakout (56 holes).

Editable PDF form

JACK	PIN	AU PIN	TYPE	USED	SIGNAL / PURPOSE	DIR	STATUS
J11	1	GND	GND				
J11	2	B3	IO N	☐			
J11	3	B9	IO N	☐			
J11	4	B15	IO N	☐			
J11	5	B21	IO N	☐			
J11	6	B27	IO N	☐			
J11	7	B33	IO - QWIIC	☐			
J11	8	B39	IO N - LCLK	☐			
J11	9	B45	IO N - GCLK	☐			
J11	10	B51	IO N MV	☐			
J11	11	B57	IO N MV	☐			
J11	12	B63	IO N MV	☐			
J11	13	B69	IO N MV	☐			
J11	14	B75	IO MV	☐			

JACK	PIN	AU PIN	TYPE	USED	SIGNAL / PURPOSE	DIR	STATUS
J12	1	GND	GND				
J12	2	B5	IO P	☐			
J12	3	B11	IO P	☐			
J12	4	B17	IO P	☐			
J12	5	B23	IO P	☐			
J12	6	B29	IO P	☐			
J12	7	B35	IO - QWIIC	☐			
J12	8	B41	IO P - LCLK	☑	1 kHz TEST	OUT	PASS
J12	9	B47	IO P - GCLK	☐			
J12	10	B53	IO P MV	☐			
J12	11	B59	IO P MV	☐			
J12	12	B65	IO P MV	☐			
J12	13	B71	IO P MV	☐			
J12	14	B77	IO P MV	☐			

JACK	PIN	AU PIN	TYPE	USED	SIGNAL / PURPOSE	DIR	STATUS
J13	1	GND	GND				
J13	2	B4	IO N	☐			
J13	3	B10	IO N	☐			
J13	4	B16	IO N	☐			
J13	5	B22	IO N	☐			
J13	6	B28	IO N	☑	UART-B /CS	OUT	PASS
J13	7	B34	IO N	☐			
J13	8	B40	IO N - LCLK	☐			
J13	9	B46	IO 1V35	☐			
J13	10	B52	IO N MV	☐			
J13	11	B58	IO N MV	☐			
J13	12	B64	IO N MV	☐			
J13	13	B70	IO N MV	☐			
J13	14	B76	IO N MV	☐			

JACK	PIN	AU PIN	TYPE	USED	SIGNAL / PURPOSE	DIR	STATUS
J14	1	GND	GND				
J14	2	B6	IO P	☐			
J14	3	B12	IO P	☐			
J14	4	B18	IO P	☐			
J14	5	B24	IO P	☑	RAM /CS	OUT	PASS
J14	6	B30	IO P	☑	ACIA /CS	OUT	PASS
J14	7	B36	IO P	☐			
J14	8	B42	IO P - LCLK	☑	ROM /CS	OUT	PASS
J14	9	B48	IO 1V35	☐			
J14	10	B54	IO P MV	☐			
J14	11	B60	IO P MV	☐			
J14	12	B66	IO P MV	☐			
J14	13	B72	IO P MV	☐			
J14	14	B78	IO P MV	☐			

DIR: IN / OUT / BIDIR. STATUS: PLANNED / WIRED / PASS. 5 V logic into FPGA I/O requires level shifting.

BR2 Physical Breakout Map - Control

The 24-hole Control area is the four 6-pin breakout headers J17-J20 from the schematic.

Editable PDF form

JACK	PIN	LABEL	MEANING / NOTES	MEASURED	STATUS
J17	**1**	**+3V3**	3.3 V supply rail		
J17	**2**	**LED0**	Au LED 0		
J17	**3**	**LED2**	Au LED 2		
J17	**4**	**RESET**	FPGA reset		
J17	**5**	**TMS**	JTAG TMS		
J17	**6**	**TDI**	JTAG data in		
J18	**1**	**VCC**	Separate VCC rail - verify voltage before use		
J18	**2**	**LED1**	Au LED 1		
J18	**3**	**LED3**	Au LED 3		
J18	**4**	**PROGRAM_B**	FPGA program/configuration		
J18	**5**	**TCK**	JTAG clock		
J18	**6**	**TDO**	JTAG data out		
J19	**1**	**DONE**	FPGA configuration done		
J19	**2**	**LED4**	Au LED 4		
J19	**3**	**LED6**	Au LED 6		
J19	**4**	**VBSEL**	Bank-voltage select		
J19	**5**	**AVP**	Analog positive input		
J19	**6**	**AREF**	Analog reference		
J20	**1**	**A1V8**	Analog 1.8 V rail		
J20	**2**	**LED5**	Au LED 5		
J20	**3**	**LED7**	Au LED 7		
J20	**4**	**+2.5V**	2.5 V rail		
J20	**5**	**AVN**	Analog negative input		
J20	**6**	**AGND**	[illegible]		

Bench note:

J17 pin 1 is explicitly +3V3. VCC is a separate rail; measure/verify it before using it as a supply.